

## ■ Week 2 · Task 4

# Linux Basics + Log Files (/var/log)

### Internship Practical Report

## 1. Introduction

This report documents practical Linux and log-analysis learning completed from a SOC analyst perspective. The work focused on command-line navigation, exploring /var/log, viewing logs, inspecting recent events, searching log data, and understanding how logs support cybersecurity investigations.

## 2. Task Objectives

Practice ls, cd, cat, tail, and grep; explore /var/log; understand failed-login searches and tail -f; complete the required Linux and log-analysis learning; and document practical evidence with screenshots.

## 3. Linux and Log Analysis Learning

The project includes evidence from Linux Fundamentals Part 1 and Intro to Logs. These learning activities established the Linux and log-analysis concepts used throughout the practical work.

## 4. Exploring Linux and /var/log

Linux navigation commands such as pwd, ls, and cd were practiced. The /var/log directory was explored because it is a major location for Linux system and application logs.

## 5. Viewing and Inspecting Logs

The cat command was used to view log-file contents, while tail was used to inspect the newest entries. The command tail -f is used for live monitoring as new log events are written.

## 6. Searching Logs

grep is used to filter large log files and locate relevant events. A common authentication investigation example is grep 'Failed' auth.log, which searches for failed-login events when that log file and those events are available.

## 7. SOC Relevance

Logs are security evidence. Analysts use them to investigate authentication activity, service events, errors, suspicious behaviour, and timelines. Parsing, normalisation, enrichment, and correlation help transform raw records into useful investigation data.

## 8. Conclusion

This task strengthened foundational Linux and log-analysis skills that support later work with SIEM platforms, centralised logging, threat hunting, and incident response.

## ■ Screenshot Evidence

The following pages contain the original screenshot files supplied for this project. Filenames have been preserved.

# Screenshot Evidence — l1.png

The screenshot shows the TryHackMe website interface. The browser address bar displays the URL: `tryhackme.com/room/linuxfundamentalspart1?utm_source=chatgpt.com`. The website's navigation bar includes links for Dashboard, Learn, Practice, and Compete, along with a Go Premium button and user statistics (34 diamonds, 0 emeralds). The main content area features the 'Linux Fundamentals (Pt1)' room header, which includes a penguin icon, a description of the room, and a 'Room progress: 0%' indicator. Below the header, there is a task section titled 'Task 1 Introduction' with a sub-section 'Set up your virtual environment'. This section contains a description of the task and a 'Lab machine' status indicator (Off) with a 'Start Lab Machine' button. A Windows watermark is visible in the bottom right corner.

tryhackme.com/room/linuxfundamentalspart1?utm\_source=chatgpt.com

TryHackMe Dashboard Learn Practice Compete

Go Premium 34 0

← Back to all walkthroughs

## Linux Fundamentals (Pt1)

Embark on the journey of learning the fundamentals of Linux. Learn to run some of the first essential commands on an interactive terminal.

15 min 1,234,201

Save Room 226 Recommendations Connectivity Options

Room progress: 0%

Task 1 Introduction

### Set up your virtual environment

To successfully complete this room, you'll need to set up your virtual environment. This involves starting the Lab Machine, ensuring you're equipped with the necessary tools and access to tackle the challenges ahead.

Lab machine Status: Off Start Lab Machine

Activate Windows  
Go to Settings to activate Windows.

## Screenshot Evidence — I2.png

# Screenshot Evidence — l3.png

The screenshot shows a web browser window at `tryhackme.com/room/linuxfundamentalspart1`. The page is titled "Room progress: 11%" and displays "Active machines information" with an IP address of `10.49.168.72`. The main content area explains the importance of the terminal in Linux and introduces basic commands. It includes two interactive boxes: one for `whoami` (tells you who you are on the system) and one for `echo` (output some specific text that is provided). A "You'll need to..." section lists two steps: 1. Use the `whoami` command to see who you are on the system. 2. Press the Enter key to run the command. On the right, a terminal window shows the following commands and output:

```
tryhackme@linux1:~$  
tryhackme@linux1:~$  
tryhackme@linux1:~$ whoami  
tryhackme  
tryhackme@linux1:~$
```

An "Activate Windows" watermark is visible in the bottom right corner of the terminal window.

# Screenshot Evidence — l4.png

tryhackme.com/room/linuxfundamentalspart1

Room progress: 11%

IP Address: 10.49.168.72

### Active machines information

Most of your time here in Cyber security, from running hacking tools to running down attackers.

A command is an instruction that we can give the computer to perform a given task. One of the first commands we can do is `whoami`. This is important in Cyber security because you will often be changing users on the machine, which determines what you can and cannot do.

`whoami` tells you who you are on the system

When you run your first command, you will see some text. We call this output. Interacting with Linux feels like a conversation. You give it an instruction, and it gives you output. We can ask Linux to output specific text for us.

`echo` output some specific text that is provided

To echo the text "TryHackMe", we can use `echo TryHackMe`. For multiple words, we will need to wrap these in quotes. For example, `echo "hello world"`. Using `echo` is just the start of a very important skill in Cyber security: being able create output and sending it somewhere. You will get to see how this skill is further used in the upcoming tasks.

**You'll need to...**

1. Use the `whoami` command to see who you are on the system.
2. Press the Enter key to run the command.
3. Now, use `echo` to output the text TryHackMe.

```
tryhackme@linux1:~$  
tryhackme@linux1:~$  
tryhackme@linux1:~$ whoami  
tryhackme  
tryhackme@linux1:~$ echo hello world  
hello world  
tryhackme@linux1:~$
```

Activate Windows  
Go to Settings to activate Windows.

linuxfundpartiv3 5min 12s

# Screenshot Evidence — I5.png

tryhackme.com/room/linuxfundamentalspart1

Room progress: 66%

IP Address: 10.49.168.72

Active machines information

4. Lost? Use `pwd` to see where we are within the system.

Why you're doing this

Answer the questions below

Run `ls` in the current folder. How many folders are there?

4

✓ Correct Answer

One of those folders contains a file. Which folder is it?

folder1

✓ Correct Answer

Use `cat` to read the file within this folder. What does it say?

password123

✓ Correct Answer

Task 1 Let the Machine Do the Searching

Task 2 Shell Operators (Combine Commands)

Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1064-aws x86\_64)

\* Documentation: <https://help.ubuntu.com>  
\* Management: <https://landscape.canonical.com>  
\* Support: <https://ubuntu.com/pro>

System information as of Tue Aug 25 05:31:15 UTC 2026

System load: 0.01 Processes: 107  
Usage of /: 22.8% of 11.56GB Users logged in: 1  
Memory usage: 15% IPv4 address for ens5: 10.49.168.72  
Swap usage: 0%

\* Ubuntu Pro delivers the most comprehensive open source security and compliance features.  
<https://ubuntu.com/aws/pro>

Expanded Security Maintenance for Applications is not enabled.  
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.  
See <https://ubuntu.com/esm> or run: `sudo pro status`

The list of available updates is more than a week old.  
To check for new updates run: `sudo apt update`  
Failed to connect to <https://changelogs.ubuntu.com/meta-release-lts>. Check your Internet connection or proxy settings

Last login: Tue Aug 25 05:31:39 2026 from 10.49.110.101  
tryhackme@linux1:~\$

Activate Windows  
Go to Settings to activate Windows.

# Screenshot Evidence — l6.png

The screenshot shows a web browser at `tryhackme.com/room/linuxfundamentalspart1`. The interface is divided into two main sections: a task area on the left and a terminal window on the right.

**Task Area (Left):**

- Active machines information:** Shows a profile icon and instructions: "2. Run `grep 'THM' access.log` to use grep to search the log file for us." and "3. Copy the `THM{ }` output that has been found."
- Why you're doing this:** A link to explain the task.
- Answer the questions below:** A section for answering questions.
- Question:** "After using `grep THM access.log`, a flag was found. What is the flag?"
- Answer:** A text input field contains `THM{ACCESS}`. Below it, a green button says "Correct Answer".
- Task 5:** "Shell Operators (Combining Commands)".
- Feedback:** A section titled "How likely are you to recommend this room to others?" with a rating scale from 1 to 10. The "Submit now" button is visible.

**Terminal Window (Right):**

```
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 5.15.0-1064-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Tue Aug 25 05:33:57 UTC 2026

System load:  0.08               Processes:    107
Usage of /:   22.8% of 11.56GB   Users logged in: 1
Memory usage: 15%               IPv4 address for ens5: 10.49.168.72
Swap usage:   0%

* Ubuntu Pro delivers the most comprehensive open source security and compliance features.

https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Tue Aug 25 05:32:09 2026 from 10.49.84.22
tryhackme@linux1:~$
```

The terminal window also shows a Windows activation watermark: "Activate Windows Go to Settings to activate Windows."



# Screenshot Evidence — I7.png

tryhackme.com/room/linuxfundamentalspart1

← Back to all walkthroughs

## Linux Fundamentals (Pt1)

Embark on the journey of learning the fundamentals of Linux. Learn to run some of the first essential commands on an interactive terminal.

📶 🟢 ⌚ 15 min 👤 1,234,203 🧑

🏆 Share your achievement Show Split View 🏠 Save Room 👍 226 Recommendations 🔗 Connectivity ⚙️ Options

Room completed: 100%

📖 Active machines information IP Address: 10.49.168.72

- Task 1 🟢 Introduction
- Task 2 🟢 Talking to Linux
- Task 3 🟢 Finding Your Way Around
- Task 4 🟢 Let the Machine Do the Searching
- Task 5 🟢 Shell Operators (Combining Commands)

How likely are you to recommend this room to others?

Activate Windows  
Go to Settings to activate Windows.

# Screenshot Evidence — l8.png

The screenshot shows the TryHackMe website interface for the 'Intro to Logs' room. The browser address bar displays the URL: `tryhackme.com/room/introtologs?utm_source=chatgpt.com`. The navigation bar includes links for Dashboard, Learn, Practice, and Compete, along with a 'Go Premium' button and user statistics (34 diamonds, 1 flame). The room title 'Intro to Logs' is prominently displayed, accompanied by a description: 'Learn the fundamentals of logging, data sources, collection methods and principles to step into the log analysis world.' Below the title, a progress bar indicates 'Room progress: 35%'. A list of tasks is shown, with Task 4 'Collection, Management, and Centralisation' selected. The main content area begins with the section 'Log Collection', which explains that log collection is an essential component of log analysis, involving the aggregation of logs from diverse sources such as servers, network devices, software, and databases. It further states that for logs to effectively represent a chronological sequence of events, it's crucial to maintain the system's time accuracy during logging, utilizing the **Network Time Protocol (NTP)** as a method to achieve this synchronisation and ensure the integrity of the timeline stored in the logs. A Windows watermark is visible in the bottom right corner of the page.

tryhackme.com/room/introtologs?utm\_source=chatgpt.com

TryHackMe Dashboard Learn Practice Compete

SOC Level 2 (Legacy) > Log Analysis > Intro to Logs

## Intro to Logs

Learn the fundamentals of logging, data sources, collection methods and principles to step into the log analysis world.

Start AttackBox Save Room 1347 Recommend Connectivity Options

Room progress: 35%

- Task 1 Introduction
- Task 2 Expanding Perspectives: Logs as Evidence of Historical Activity
- Task 3 Types, Formats, and Standards
- Task 4 Collection, Management, and Centralisation

### Log Collection

Log collection is an essential component of log analysis, involving the aggregation of logs from diverse sources such as servers, network devices, software, and databases.

For logs to effectively represent a chronological sequence of events, it's crucial to maintain the system's time accuracy during logging. Utilising the **Network Time Protocol (NTP)** is a method to achieve this synchronisation and ensure the integrity of the timeline stored in the logs.

Activate Windows  
Go to Settings to activate Windows.

# Screenshot Evidence — I9.png

The screenshot displays a web browser window at `tryhackme.com/room/introtologs?utm_source=chatgpt.com`. The left sidebar shows the room progress at 79% and a list of tasks. Task 7, 'Conclusion', is the active task, indicating completion of the 'Intro to Logs' room. The main content area shows a congratulatory message and a summary of learning objectives.

**Active machines information**  
IP Address: 10.48.153.74

**Task 1** Introduction

**Task 2** Expanding Perspectives: Logs as Evidence of Historical Activity

**Task 3** Types, Formats, and Standards

**Task 4** Collection, Management, and Centralisation

**Task 5** Storage, Retention, and Deletion

**Task 6** Hands-on Exercise: Log analysis process, tools, and techniques

**Task 7** Conclusion

Congratulations! You've completed the Intro to Logs room.

In summary, we were able to learn and perform the following:

- The significance of logs as records of past activities; essential for pinpointing and tackling threats.
- Delve into an array of logs, their creation techniques, and the methods of gathering them from diverse systems.
- Review the results from analysing logs in the realms of detection engineering and incident handling.
- Acquire practical skills in identifying and countering adversaries via log analysis.

**Terminal Window:**

```
root@ip-10-48-153-74:~#  
root@ip-10-48-153-74:~# ls -la /etc/logrotate.d/  
total 112  
drwxr-xr-x 2 root root 4096 Aug 20 19:44 .  
drwxr-xr-x 195 root root 12288 Aug 26 07:10 ..  
-rw-r--r-- 1 root root 120 Feb 5 2024 alternatives  
-rw-r--r-- 1 root root 397 Mar 18 2024 apache2  
-rw-r--r-- 1 root root 126 Apr 22 2022 apport  
-rw-r--r-- 1 root root 173 Mar 22 2024 apt  
-rw-r--r-- 1 root root 174 Aug 20 05:54 attackbox-vpn  
-rw-r--r-- 1 root root 1154 Aug 20 19:49 attackbox-vpn-ensure  
-rw-r--r-- 1 root root 91 Jan 4 2024 bootlog  
-rw-r--r-- 1 root root 130 Oct 14 2019 btpp  
-rw-r--r-- 1 root root 160 Oct 2 2024 chrony  
-rw-r--r-- 1 root root 144 Aug 12 2025 cloud-init  
-rw-r--r-- 1 root root 181 Feb 28 2024 cups-daemon  
-rw-r--r-- 1 root root 112 Feb 5 2024 dpkg  
-rw-r--r-- 1 root root 125 Aug 14 2019 lightdm  
-rw-r--r-- 1 root root 94 Aug 18 2022 ppp  
-rw-r--r-- 1 root root 248 Mar 22 2024 rsyslog  
-rw-r--r-- 1 root root 132 Sep 10 2020 sane-utils  
-rw-r--r-- 1 root root 677 Oct 4 2023 speech-dispatcher  
-rw-r--r-- 1 root root 174 Apr 5 2024 sssd-common  
-rw-r--r-- 1 root root 246 Feb 15 2024 stunnel4
```

Activate Windows  
Go to Settings to activate Windows.

THM AttackBox  
Persistence: Off 0min 12s

# Screenshot Evidence — l10.png

tryhackme.com/room/introtologs/congratulations?step=records

Room completed: 100%

```
ce/gitlab-instance-2fdab3a4/pat-api-key" "Mozilla/5.0 (X11; Linux x86_64; rv:102.0) Gecko/20100101 Firefox/102.0" -"
[14/Aug/2023:21:48:06] --- /var/log/gitlab/nginx/access.log --- "34.253.159.159 - - [14/Aug/2023:21:48:06] "GET /gitlab-instance-2fdab3a4/pat-api-key/-/blob/main/README.md?format=json&viewer=rich HTTP/1.0" 200 18904 "http://gitlab-waifrispend.finance/gitlab-instance-2fdab3a4/pat-api-key" "Mozilla/5.0 (X11; Linux x86_64; rv:102.0) Gecko/20100101 Firefox/102.0" -"
[14/Aug/2023:21:48:06] --- /var/log/gitlab/nginx/access.log --- "34.253.159.159 - - [14/Aug/2023:21:48:06] "GET /gitlab-instance-2fdab3a4/pat-api-key HTTP/1.0" 200 98890 "" "Mozilla/5.0 (X11; Linux x86_64; rv:102.0) Gecko/20100101 Firefox/102.0" -"
[14/Aug/2023:21:48:06] --- /var/log/gitlab/nginx/access.log --- "34.253.159.159 - - [14/Aug/2023:21:48:06] "GET /gitlab-instance-2fdab3a4/pat-api-key/-/refs/main/logs_tree?format=json&offset=0 HTTP/1.0" 200 531 "http://gitlab-waifrispend.finance/gitlab-instance-2fdab3a4/pat-api-key" "Mozilla/5.0 (X11; Linux x86_64; rv:102.0) Gecko/20100101 Firefox/102.0" -"
[14/Aug/2023:21:49:01] --- /var/log/websrv-02/rsyslog_cron.log --- "Aug 14 21:49:01 WEBSRV-02 CRON[30173]: (r"
```

NOTE: You can access the URL using the AttackBox or VM browser. However, please be aware that Firefox on the VM may take a few minutes to boot up.

Answer the questions below

Upon accessing the log viewer URL for unparsed raw log files, what error does "/var/log/websrv-02/rsyslog\_cron.log" show when selecting the different filters?

No date field

✓ Correct Answer

What is the process of standardising parsed data into a more easily readable and query-able format?

Normalisation

✓ Correct Answer

What is the process of consolidating normalised logs to enhance the analysis of activities related to a specific IP address?

Enrichment

✓ Correct Answer

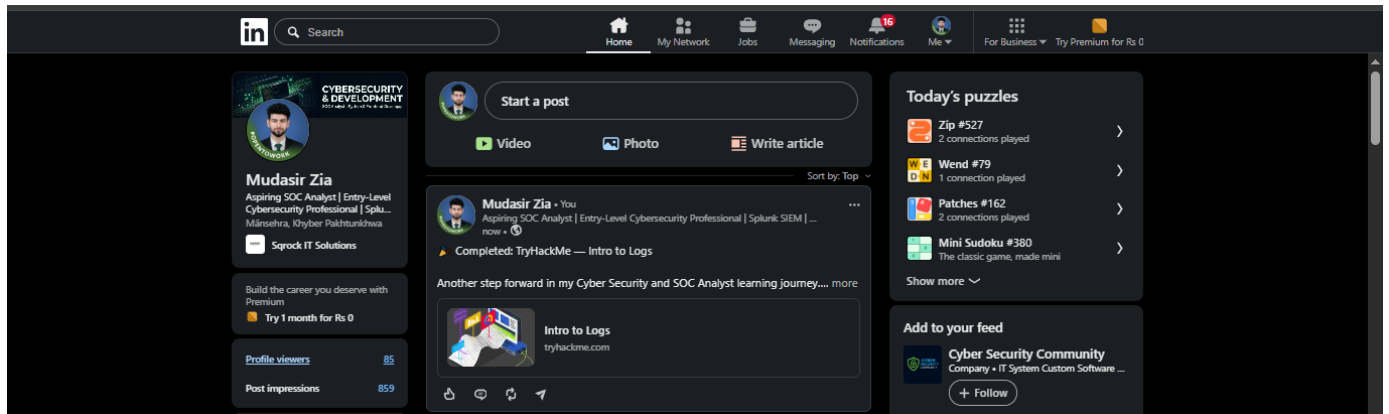
Task 7 Conclusion

How likely are you to recommend this room to others?

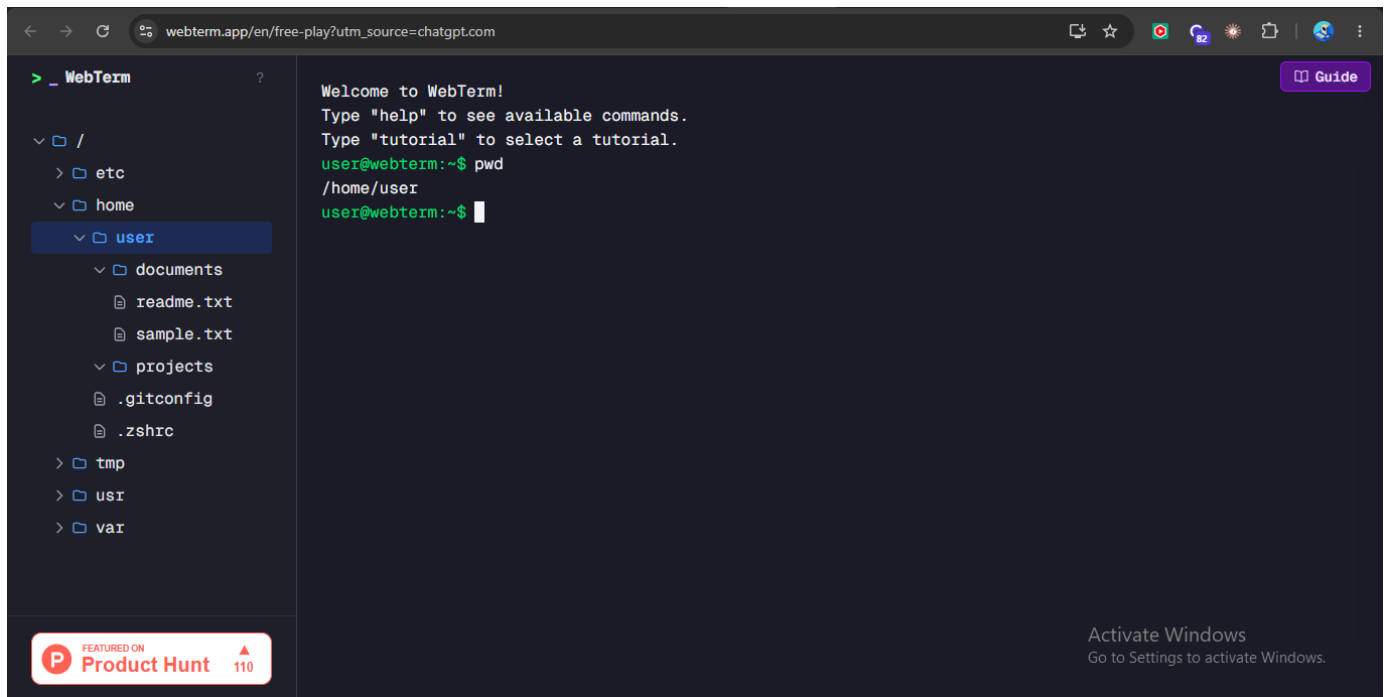
Activate Windows

Go to Settings to activate Windows.

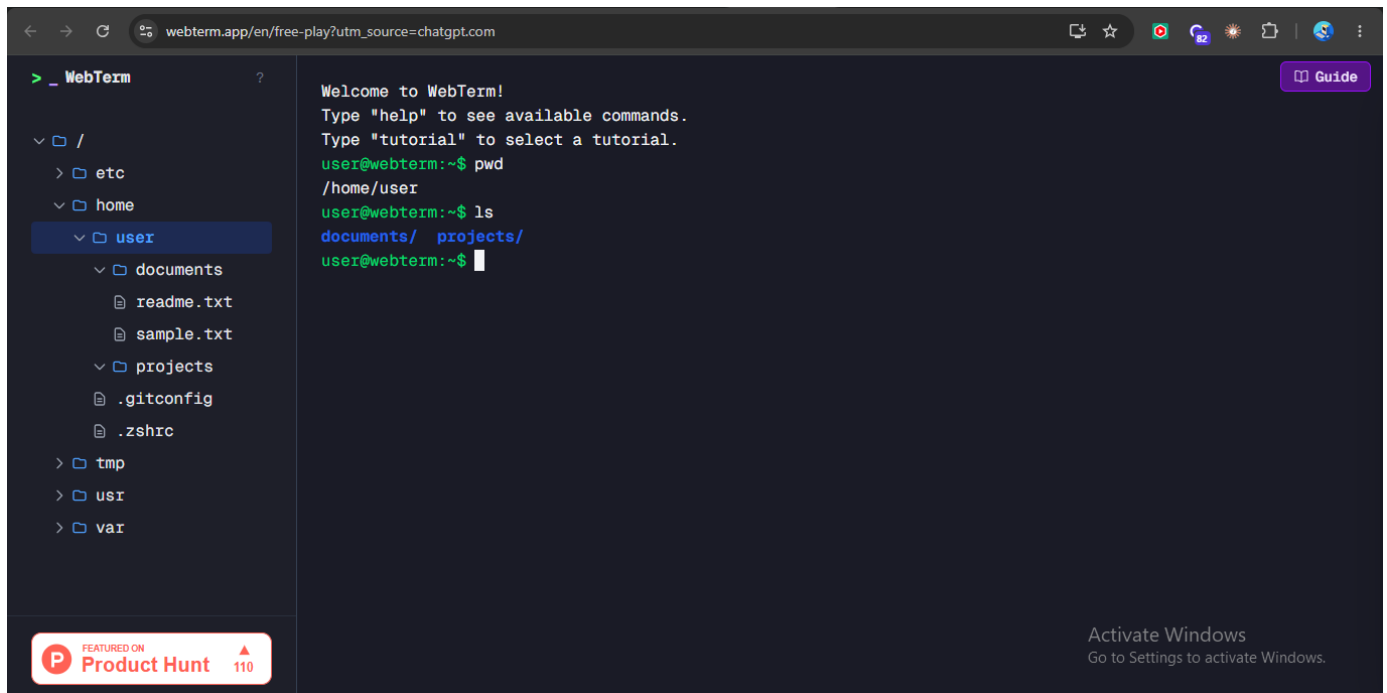
# Screenshot Evidence — l11.png



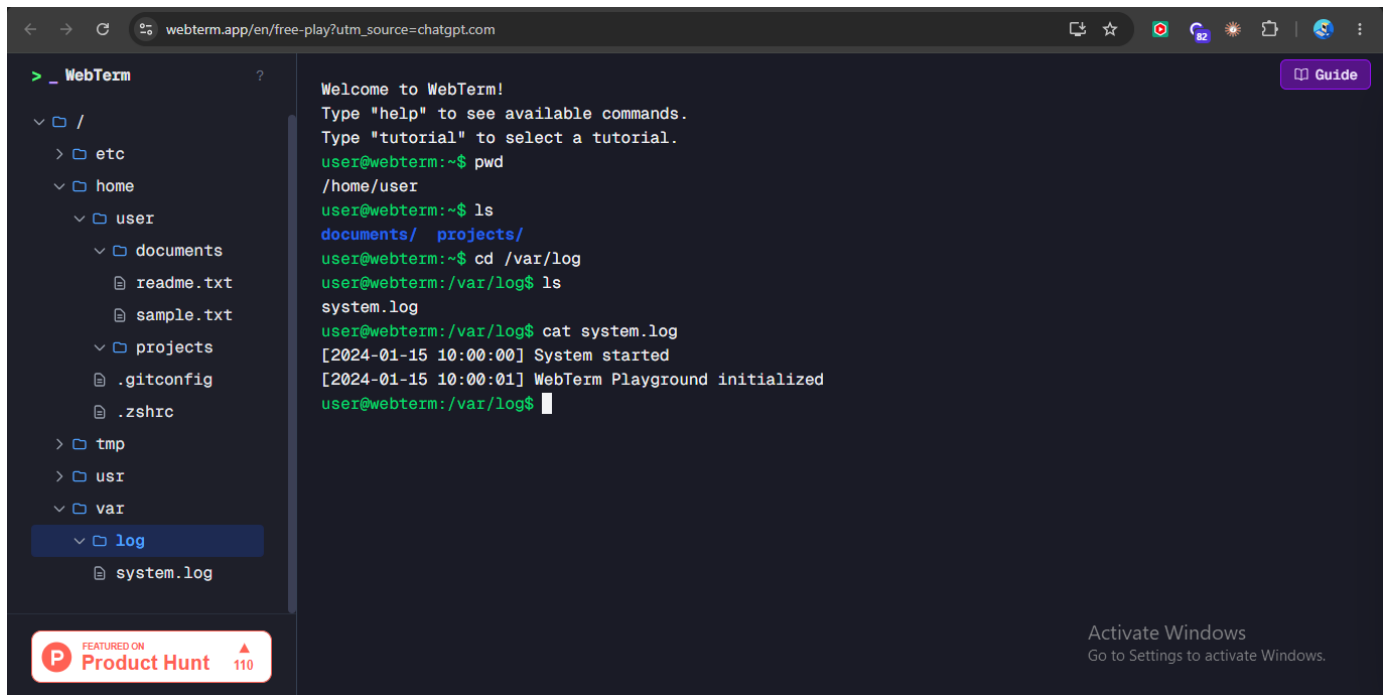
## Screenshot Evidence — l12.png



## Screenshot Evidence — l13.png

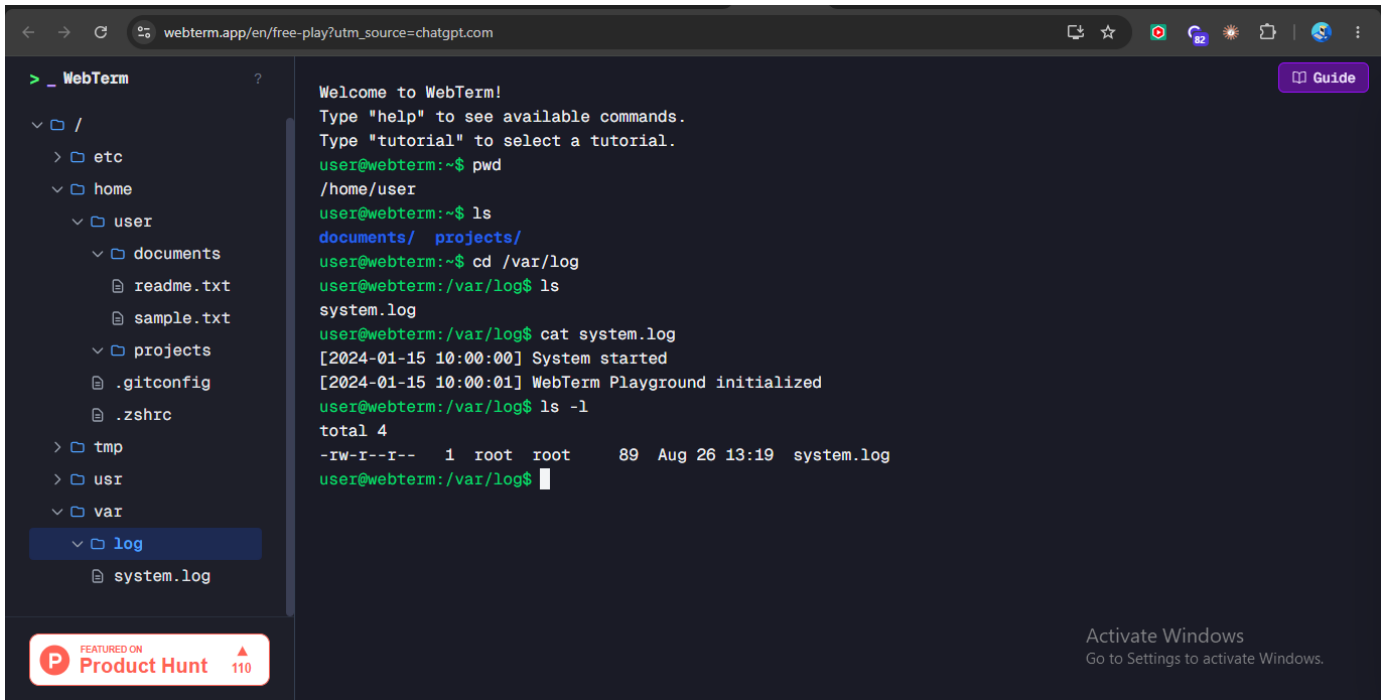


## Screenshot Evidence — l14.png

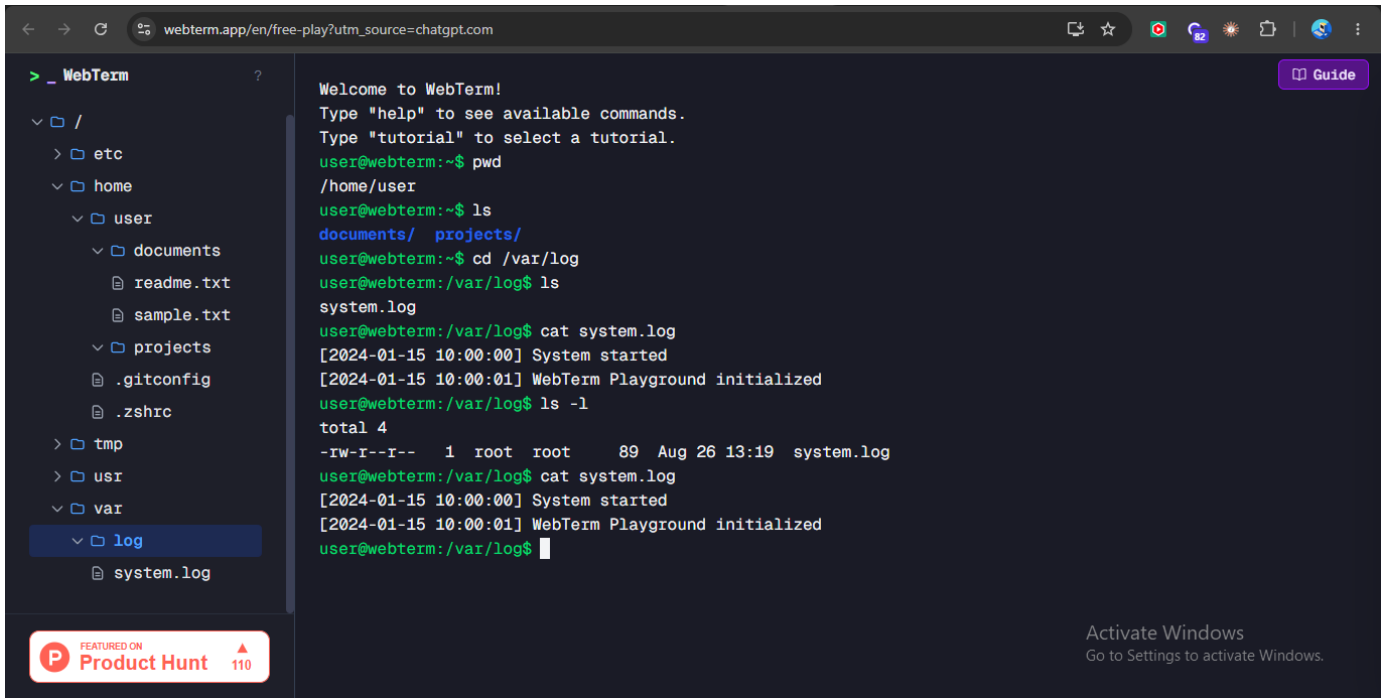




## Screenshot Evidence — I15.png



## Screenshot Evidence — I16.png



## Screenshot Evidence — l17.png

```
<  →  webterm.app/en/free-play?utm_source=chatgpt.com  ☆  🌐  🔄  📄  📁  📂  📅  📆  📇  📈  📉  📊  📋  📌  📍  📎  📏  📐  📑  📒  📓  📔  📕  📖  📗  📙  📚  📛  📜  📝  📞  📟  📠  📡  📢  📣  📤  📥  📦  📧  📨  📩  📪  📫  📬  📭  📮  📯  📰  📱  📲  📳  📴  📵  📶  📷  📸  📹  📺  📻  📼  📽  📾  📿  📠  📡  📢  📣  📤  📥  📦  📧  📨  📩  📪  📫  📬  📭  📮  📯  📰  📱  📲  📳  📴  📵  📶  📷  📸  📹  📺  📻  📼  📽  📾  📿

> _ WebTerm ?
  ▾ /
    ▸ etc
    ▾ home
      ▾ user
        ▾ documents
          📄 readme.txt
          📄 sample.txt
        ▾ projects
          📄 .gitconfig
          📄 .zshrc
      ▸ tmp
      ▸ usr
      ▾ var
        ▾ log
          📄 system.log

type "help" to see available commands.
Type "tutorial" to select a tutorial.
user@webterm:~$ pwd
/home/user
user@webterm:~$ ls
documents/  projects/
user@webterm:~$ cd /var/log
user@webterm:/var/log$ ls
system.log
user@webterm:/var/log$ cat system.log
[2024-01-15 10:00:00] System started
[2024-01-15 10:00:01] WebTerm Playground initialized
user@webterm:/var/log$ ls -l
total 4
-rw-r--r--  1 root root   89 Aug 26 13:19 system.log
user@webterm:/var/log$ cat system.log
[2024-01-15 10:00:00] System started
[2024-01-15 10:00:01] WebTerm Playground initialized
user@webterm:/var/log$ tail syslog
tail: cannot open 'syslog' for reading: No such file or directory
user@webterm:/var/log$ ls
system.log
user@webterm:/var/log$ tail system.log
[2024-01-15 10:00:00] System started
[2024-01-15 10:00:01] WebTerm Playground initialized
user@webterm:/var/log$
```

Activate Windows  
Go to Settings to activate Windows.

## **Screenshot Evidence — I18.png**